Ex-11
8/10/2024

ROUTING

11)a) STATIC ROUTING

AIM:-

→ To simulate and configure static routing using Cisco Packet Tracer.

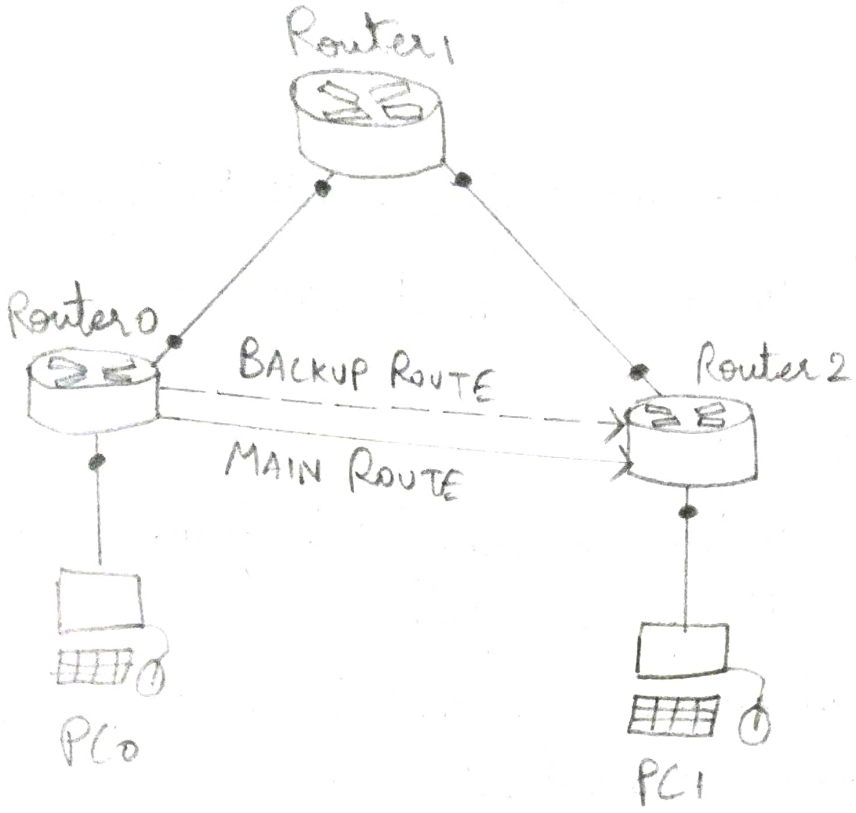
OBSERVATION:-

1) Static routing involves manually adding routes to a router's routing table.
2) Routers automatically know the networks directly connected to their interfaces.
3) Routes are manually added for networks that are available through other routers.
4) Both main and backup routes can be configured for a destination.
5) Administrative distance (AD) is used to select the preferred static route
6) The routing table is checked using:
   show ip route static
7) Connectivity and routing paths are verified using ping and tracert.
8) When the main route fails, the backup route can be used for forwarding packets.

PC-PT
PC1
PC-PT
PC0
2960-24TT
Switch0
2911
Router0
2911
Router1
2960-24TT
Switch1
PC-PT
PC2

C. ASHWIN
BE (CS) III yr

DIAGRAM:-



RESULT:-

Thus static routes were successfully configured and verified, including the use of a backup route when the main route becomes unavailable.

11) b)
8/10/2026

DYNAMIC ROUTING – RIP

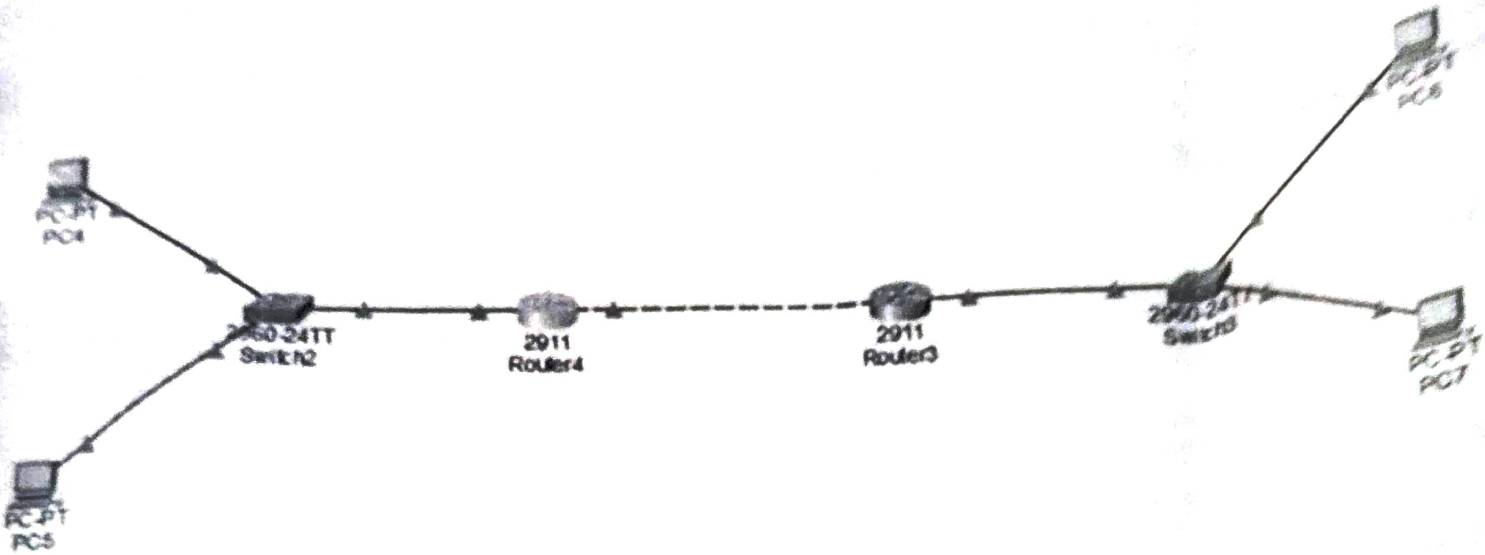
AIM:-

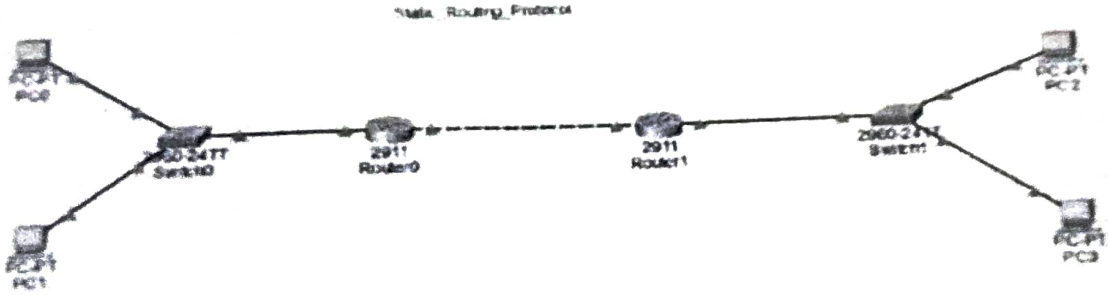
To simulate and configure RIP (Routing Information Protocol) using Cisco Packet Tracer.

OBSERVATION:-

1) Three routers are connected to form a network with multiple possible paths.
2) IP addresses are assigned to the PCs, FastEthernet interfaces and serial interfaces.
3) RIP is enabled using the router rip command
4) Directly connected networks are advertised using the network command
5) RIP automatically exchanges routing information between routers.
6) RIP selects the route based on the lowest hop count.
7) The ping command is used to test connectivity between PC0 and PC1.
8) The tracert command is used to verify the path followed by packets.
9) If the selected route fails, RIP automatically chooses another available route.

RIP_Dynamic_Routing_Protocol





RIP_Dynamic_Routing_Protocol

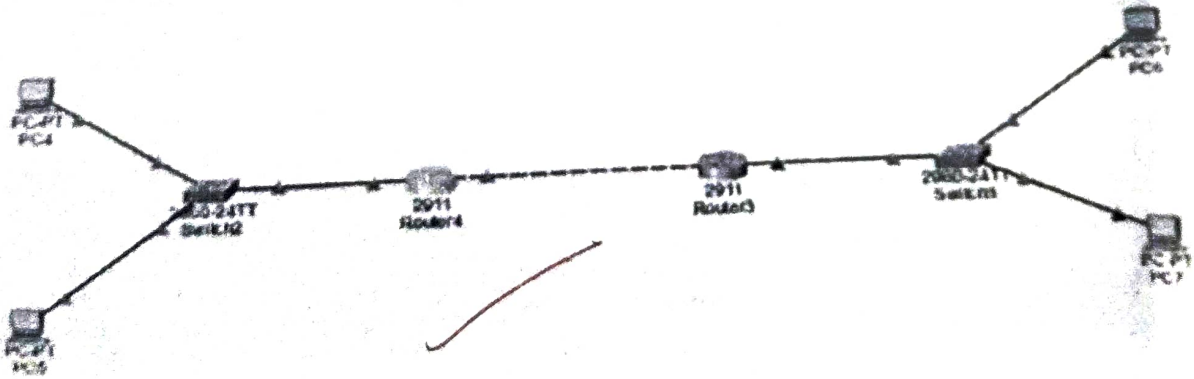
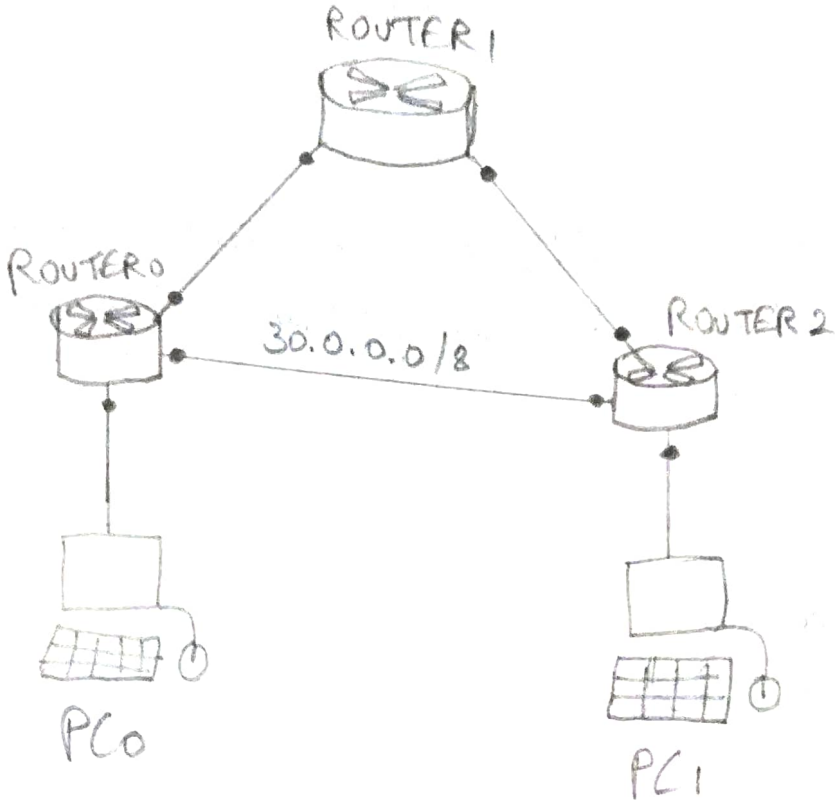


DIAGRAM:-



1)

RESULT:-

Thus RIP was successfully configured and dynamic routing was verified. The router automatically selected the available route with the lower hop count and rerouted traffic when the selected route failed.